

CS5260 Project Proposal

Problem Statement

Researchers face overwhelming information when navigating the vast amount of academic literature. Summarizing papers, identifying key contributions, and understanding connections between works is time-consuming. We propose an AI tool that automates paper summarization, generates interactive concept maps, and visualizes research trends. This will enable faster insight extraction and collaborative knowledge discovery.

This project will approach this problem by developing an AI system as described:

1. Academic Paper Summarization and Visualisation Tool
 - AI system that reads research papers and automatically extracts key points
 - Generate interactive diagrams/concept maps about key points
 - Users can annotate and comment on research landscape manually

Methodology

1. Data Collection: Gather research papers via academic APIs (e.g., arXiv, Semantic Scholar) and web scraping where necessary.
2. Summarization: Fine-tune transformer models (explore using Google PEGASUS, Huggingface BLOOM)
3. Visualisation: AI Tools to generate interactive diagrams and concept maps (spaCy for NER extraction and PyVis for diagrams/concept maps)
4. Annotation: Huggingface's Gradio Interactive dashboard

Expected Outcomes

Demo - Web platform with summarization + visualization

Video - A walkthrough of the tool's capabilities and use cases

Team Members

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